Week 8: Rook race and Nim

Adult guide built from the problem files · Unpiloted.

The answers, hints and routes are in the adult guide, week-08-facilitator.pdf, whose source is guide-src/facilitator.tex. First route = the guide's core problems.

Problems and routes

Problem	K–1	2–3	4–5	route
Two 3-by-3 boards: first or second?	P1			first
Four 4-by-4 boards: first or second?	P2			first
The winning move on four 5-by-5 boards	P3			return
Every square to move to on a 6-by-6 board	P4			return
Two piles: first or second?	P5			first
Two piles of 7: show you can always win	P6			first
The 8-by-8 board from the far corner	P7			return
Three piles: first or second?	P8			return
Four starts on a 5-by-5 board		P1		first
Colour the 8-by-8 board		P2		first
Explain the second player's win from the corner		P3		first
Two piles: first or second?		P4		first
The board as two piles		P5		first
Piles of 52 and 37		P6		return
Small three-pile games		P7		return
Every second-player start with piles of 1 to 3		P8		return
Every second-player start with piles of 1 to 6		P9		return
Colour the 8-by-8 board and explain the red squares			P1	first
The board as two piles; 23 and 17, 30 and 30			P2	return
Six three-pile starts			P3	first
Every second-player start with piles of 1 to 7			P4	first
A rule from stacks of 1, 2, 4, 8			P5	first
Who wins, and a winning first move			P6	first
Why the stack rule always works			P7	return
The queen game on the 8-by-8 board			P8	return
The queen game on a 20-by-20 board			P9	return

First route. K–1 P1–P2, P5–P6; Grades 2–3 P1–P5; Grades 4–5 P1, P3–P6. **Return visits.** K–1 P3–P4, P7–P8; Grades 2–3 P6–P9; Grades 4–5 P2, P7–P9.

Key, one entry per problem

These problem files hold no keys; the week's adult guide has them.